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Processando config 14, seed 1, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

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*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

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smultin_MultinomialBitsOver test:

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N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE

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Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = 1 / 32768
EColl = n^2 / (2k) = 128
Hashing = TRUE

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Collision test
CollisionOver: density = n / k = 1 / 32768.00
Expected number of collisions = Mu = 128.00

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Results of CollisionOver test:

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POISSON approximation :
Expected number of collisions = N*Mu : 256.00
Observed number of collisions : 267
p-value of test : 0.25

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Total number of cells containing j balls

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j = 0 : 549739036939
j = 1 : 16776682
j = 2 : 267
j = 3 : 0
j = 4 : 0
j = 5 : 0

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CPU time used : 00:00:04.49

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Generator state:
16777216 bits have been read.

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*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 16.00
p-value of test : 0.86

CPU time used : 00:00:00.06

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.06

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : -0.79
p-value of test : 0.79

CPU time used : 00:00:00.01

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 8.41
p-value of test : 0.30

Normal statistic for number of jumps : -1.73
p-value of test : 0.96

CPU time used : 00:00:00.18

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : -0.48
p-value of test : 0.68

CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : 0.38
p-value of test : 0.35

CPU time used : 00:00:00.19

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.014
p-value of test : 0.81

Kolmogorov-Smirnov- statistic = D- : 0.029
p-value of test : 0.42

Anderson-Darling statistic = A2 : 0.50
p-value of test : 0.75

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 5.13
p-value of test : 0.16

Global longest run of 1 : 23.00
p-value of test : 0.50

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 106.39
p-value of test : 0.18

CPU time used : 00:00:00.01

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 15.36
p-value of test : 0.85

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -2.30
p-value of test : 0.99

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -0.091
p-value of test : 0.54

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 0.98
p-value of test : 0.16

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 198.86
p-value of test : 0.12

CPU time used : 00:00:00.01

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 293.78
p-value of test : 0.67

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 465.51
p-value of test : 0.77

CPU time used : 00:00:00.01

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

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Normal statistic          :    0.58
p-value of test          :    0.28
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CPU time used            : 00:00:00.00
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Generator state:
269499904 bits have been read.
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin
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sstring_AutoCor test:
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N = 1, n = 16777154, r = 0, s = 32, d = 2
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```

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Normal statistic          :   -0.89
p-value of test          :    0.81
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CPU time used            : 00:00:00.01
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Generator state:
286277088 bits have been read.
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*****
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HOST = matheus-IdeaPad-3-15ALC6, Linux
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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin
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sstring_Run test:
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N = 1, n = 3355443, r = 0, s = 32
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Total number of 1 runs: 3355448
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Number of degrees of freedom : 36
Chi2 statistic for number of runs : 38.26
p-value of test : 0.37
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Total number of bits: 13423584
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Normal statistic for number of bits : 0.35
p-value of test : 0.36
```

CPU time used : 00:00:00.05

Generator state:
299700672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 4.11
p-value of test : 0.25

CPU time used : 00:00:00.06

Generator state:
316477888 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.07

Generator state:
333169088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	31.57
p-value of test	:	0.88

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	42.98
p-value of test	:	0.43

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	55.58
p-value of test	:	0.76

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	30.39
p-value of test	:	0.73

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	29.04
p-value of test	:	0.14

CPU time used : 00:00:00.06

Generator state:
349946304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	102.30
p-value of test	:	0.26

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	94.12
p-value of test	:	0.48

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	489.35
p-value of test	:	0.76

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	60.14
p-value of test	:	0.9935

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	44.79
p-value of test	:	0.68

CPU time used : 00:00:00.05

Generator state:

366723520 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	161.19
p-value of test	:	0.03

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	109.67
p-value of test	:	0.91

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	163.34
p-value of test	:	0.43

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	166.21
p-value of test	:	0.02

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	93.16
p-value of test	:	0.36

CPU time used : 00:00:00.02

Generator state:

383500320 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.13
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
22 MatrixRank, 320 x 320	eps
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All other tests were passed